



Has BESS Stolen the Spotlight from EVs?

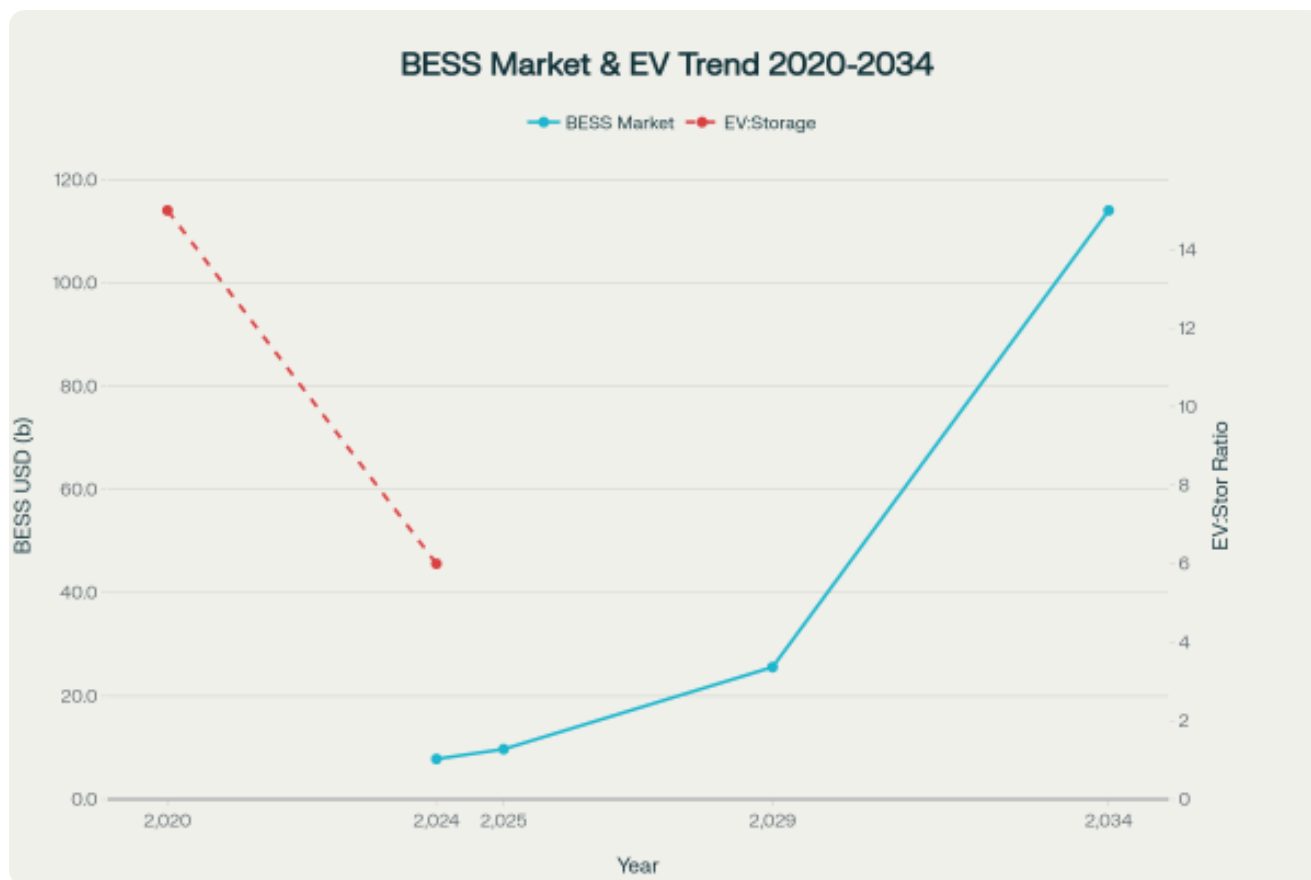


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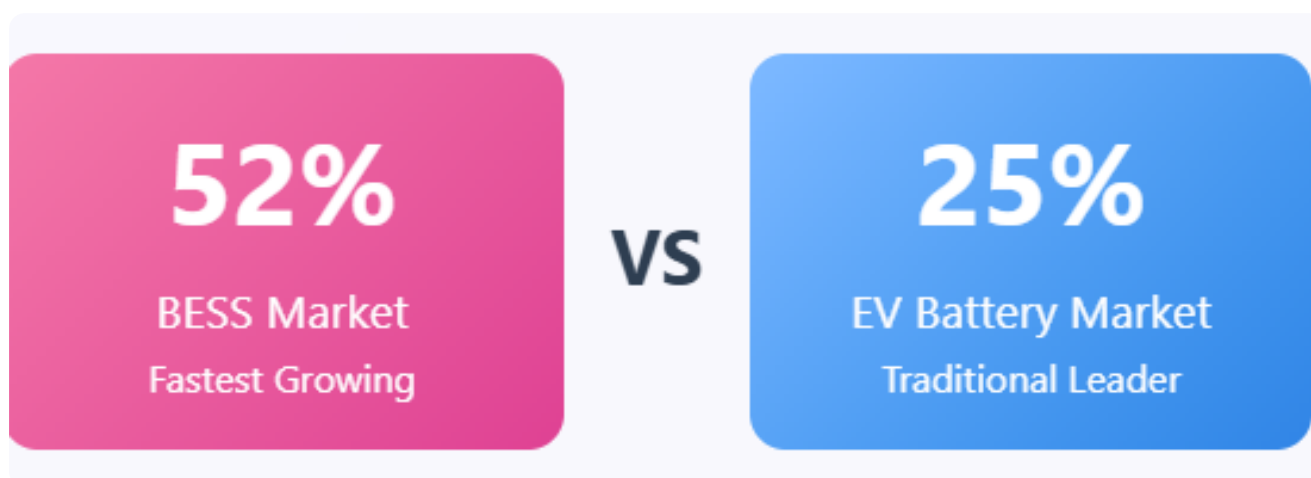
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The energy storage revolution is quietly reshaping the battery industry landscape, with Battery Energy Storage Systems (BESS) emerging as the fastest-growing segment in the global battery market. While electric vehicles (EVs) dominated headlines for over a decade, a new narrative is unfolding where stationary energy storage is capturing unprecedented attention from investors, policymakers, and industry leaders. BESS Market Growth vs EV Battery Demand Dynamics (2020-2034) BESS Market Growth vs EV Battery Demand Dynamics (2020-2034)

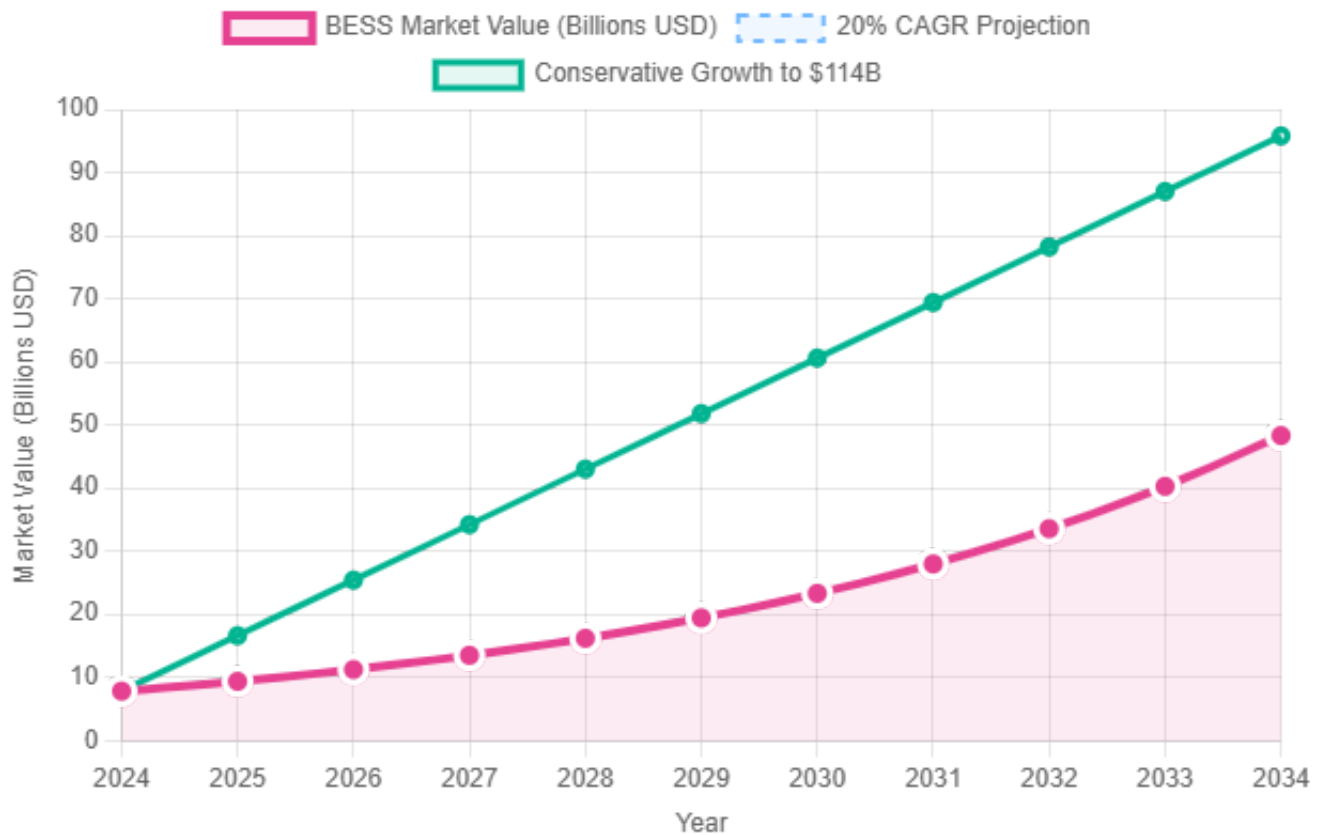


The Numbers Don't Lie: BESS Outpaces EV Growth

The data reveals a dramatic shift in battery market dynamics. **BESS has been the fastest-growing battery demand market globally for three consecutive years.** In 2024, the BESS market grew by an impressive **52%** compared to **25% growth in the EV battery market.** This represents a fundamental change from the traditional EV-dominated battery landscape.



The global BESS market has experienced explosive growth, expanding from **\$7.8 billion in 2024 to a projected \$114.05 billion by 2034**, representing a compound annual growth rate (CAGR) of nearly 20%. Various market analyses project even higher growth rates, with some forecasting CAGRs of **26.9% to 32.69%** depending on market segments and regional focus.



Changing Market Share Dynamics

Perhaps most telling is the shift in battery demand allocation. **BESS now accounts for 13% of total global battery demand in 2024, up from just 6% in 2020.** The ratio of EV battery demand to stationary battery demand has dramatically compressed from **15-to-1 in 2020 to 6-to-1 in 2024**, indicating that stationary storage is becoming a material component of the global battery ecosystem.

This trend is particularly evident in markets like China, where **energy storage applications overtook consumer electronics as the second-largest application for battery production in 2024.** The Chinese market exemplifies this shift, with the country accounting for approximately **two-thirds of installed BESS capacity globally.**

Investment Patterns Reflect the Shift

The investment landscape tells a compelling story of BESS gaining financial prominence. **Total funding for energy storage reached \$17.6 billion in the first nine months of 2024, representing a 15% increase.** While venture capital funding for energy storage companies decreased by 69% to \$2.7 billion, this was offset by a massive 125% surge in debt and public market financing to \$15 billion.



Government support has been particularly robust, with India's **Viability Gap Funding (VGF) scheme** providing \$1.1 billion for BESS development, and the scheme recently expanded to support an additional 30GWh of projects with \$631.5 million in funding. The Indian government expects this to attract \$33 billion in private investment.

The Software Revolution Within BESS

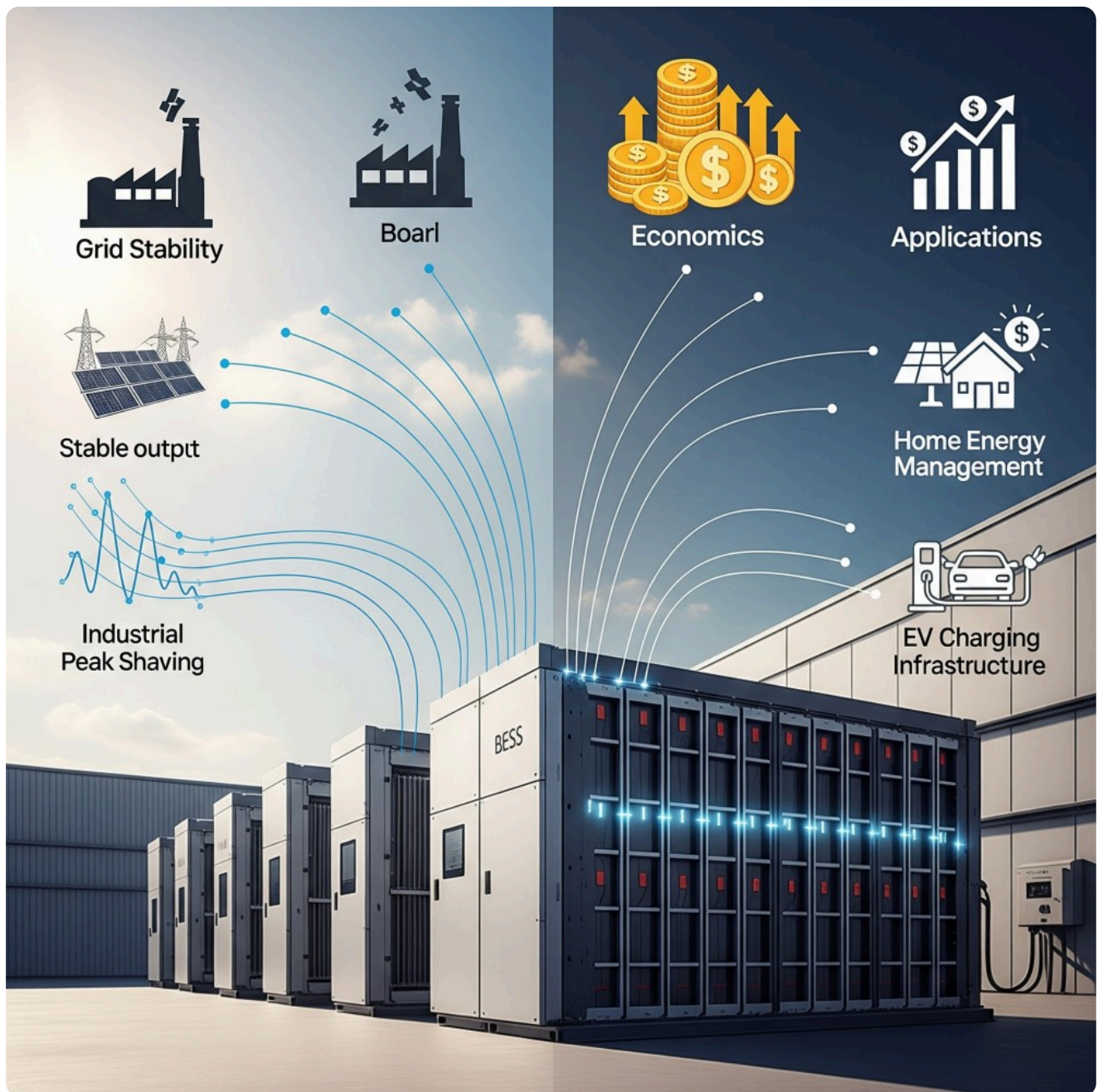
A particularly interesting development is the emergence of BESS software as a high-value investment category. **The BESS software market is projected to reach \$150 billion by 2030**, with venture capital firms increasingly looking beyond hardware to the growing software layer. **Soft costs now account for over 40% of BESS unit expenses**, making software solutions essential for optimizing operations and maximizing returns.

Why BESS is Capturing Attention

Several factors explain BESS's remarkable ascent:

- **Grid Stability Requirements:** As renewable energy penetration increases globally, **BESS provides critical grid stabilization services**. The technology addresses the intermittency challenges of solar and wind power, making renewables more viable and reliable.
- **Economic Viability:** **Battery costs have declined by around 80% over the past decade**, making BESS projects economically attractive. In India, **merchant BESS operations turned profitable for the first time in 2024**, with new projects potentially delivering 17% internal rates of return.
- **Policy Support:** Governments worldwide are implementing supportive frameworks. **Europe added 21.9 GWh of BESS capacity in 2024**, while the **US BESS market delivered 60% growth in 2024**, rising from 6 GW to 10 GW of installations.
- **Diverse Applications:** Unlike EVs, which serve primarily transportation needs, BESS offers multiple value streams including **energy arbitrage, frequency regulation, peak shaving, and**

backup power.



EV Market Challenges Creating Opportunities for BESS

While the EV market continues growing, it faces several headwinds that have created space for BESS to shine:

Slower Growth Rates: EV sales growth has decelerated in many regions, with **battery manufacturers like LG Energy Solutions and Samsung SDI posting falling quarterly revenues**. Even CATL, the world's largest battery maker, reported its first quarterly profit drop.

Infrastructure Constraints: The EV industry faces persistent challenges including **charging infrastructure limitations, range anxiety, and supply chain disruptions**. These issues have slowed adoption rates in key markets.

Market Saturation Concerns: In mature EV markets, growth rates are normalizing, creating opportunities for other battery applications to capture greater market share.

EV MARKET CHALLENGES CREATE OPPORTUNITIES FOR BATTERY ENERGY STORAGE SYSTEMS (BESS)

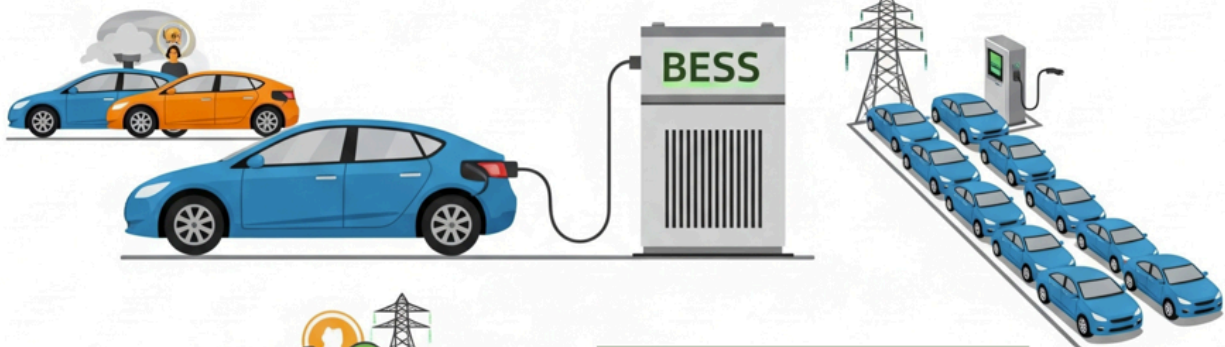
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Regional Variations and Opportunities

The BESS boom is not uniform globally. **China leads with 215.5 GWh of installed capacity and an ambitious 505.6 GWh project pipeline.** The **US follows with 82.1 GWh installed and 162.5 GWh planned.**

Emerging markets show particularly strong growth potential. **Canada is projected to be the fastest-growing BESS market through 2027, with capacity increasing from 0.3 GWh to 18.3 GWh.** Saudi Arabia's capacity could increase 24-fold to reach 32.4 GWh.

The Symbiotic Relationship

Rather than viewing BESS and EVs as competitors, the reality is more nuanced. **The growth of EV adoption is actually creating demand for BESS solutions, particularly in charging infrastructure.**

EV charging stations increasingly rely on BESS for **peak shaving, load management, and cost optimization.**



Moreover, **second-life EV batteries** are finding new applications in BESS installations, creating a circular economy that benefits both sectors. **Used EV batteries still maintain 70-80% of their capacity**, making them suitable for stationary storage applications.

Looking Ahead: A New Energy Storage Era

The evidence suggests that BESS hasn't simply stolen the spotlight from EVs—it has earned it through exceptional growth, compelling economics, and critical infrastructure value. **Global energy storage installations are expected to rise 61% in 2025, while annual battery storage installations are predicted to exceed 400 GWh by 2030.**

The energy storage market is projected to grow to \$5.12 trillion at a CAGR of 21.7% by 2034, representing nearly seven times its current value. This growth trajectory positions BESS as not just a supporting technology for renewables, but as a fundamental pillar of the future energy system.

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